Quantum Mechanics II — Exercise Sheet 8

Oscillator states and related models

Variant: Solutions

Exercise 1 (A coherent superposition with even number). For $\alpha \neq 0$ define $\Phi = C(\psi_\alpha + \psi_{-\alpha})$, where $C > 0$ normalizes the vector. Using the coherent-state expansion, compute C, all number probabilities and $\langle H \rangle$. Give the limit as $\alpha \to 0$.

Solution. Write $r = |\alpha|^2$. Adding the two number expansions cancels every odd power of α. Since $\langle \psi_\alpha, \psi_{-\alpha} \rangle = e^{-2r}$, normalization gives

$$C = [2(1 + e^{-2r})]^{-1/2}, \qquad p_{2j} = \frac{r^{2j}}{(2j)! \cosh r}, \qquad p_{2j+1} = 0.$$

Differentiating the power series for $\cosh r$ gives $\langle N \rangle = r \tanh r$. Thus

$$\langle H \rangle = \hbar\nu \left(\frac{1}{2} + r \tanh r \right).$$

As $\alpha \to 0$ the vector tends to h_0, with energy $\hbar\nu/2$.

Exercise 2 (A bounded perturbation of two levels). Let H be the oscillator Hamiltonian and let

$$B = \lambda(|h_0\rangle\langle h_2| + |h_2\rangle\langle h_0|), \qquad \lambda \in \mathbb{R}.$$

Diagonalize $H+B$ on $\operatorname{span}\{h_0, h_2\}$ and describe its action on the other number states. Explain why the domain is $D(H)$ and why this is an exact finite-dimensional reduction.

Solution. On the indicated invariant subspace the matrix is

$$\begin{pmatrix} \hbar\nu/2 & \lambda \\ \lambda & 5\hbar\nu/2 \end{pmatrix}.$$

Its eigenvalues are

$$E_\pm = \frac{3}{2}\hbar\nu \pm \sqrt{(\hbar\nu)^2 + \lambda^2}.$$

For $\lambda \neq 0$ an eigenvector has coordinates proportional to $(\lambda, E_\pm - \hbar\nu/2)$. For $\lambda = 0$ the original vectors h_0 and h_2 are eigenvectors. All h_n with $n \neq 0, 2$ retain their old energies. Diagonalize this finite block and keep every other number eigenvector. The diagonal-series construction of self-adjoint operators from Lecture 1 applies in this complete orthonormal basis. Only two coordinates have changed, so the square-summability condition for the weighted coefficients is unchanged and the domain remains $D(H)$. There is no coupling to the complementary subspace, which makes the reduction exact.

Exercise 3 (Landau levels from an oscillator). For a particle of nonzero charge e in a nonzero constant magnetic field B, take as given that a partial Fourier transform in y reduces the planar Hamiltonian to

$$H_k = \frac{P_x^2}{2m} + \frac{(\hbar k - eBx)^2}{2m}, \qquad k \in \mathbb{R}.$$

(i) Find its oscillator centre, frequency and levels. (ii) Explain why the levels do not depend on k and why a fixed-k plane wave is not a normalized state on the whole plane.

Solution. Completing the square writes the potential as $m\nu_c^2(x-x_k)^2/2$, where

$$x_k = \frac{\hbar k}{eB}, \qquad \nu_c = \frac{|eB|}{m}.$$

A translation in x is unitary and does not change the oscillator levels. Therefore each reduced Hamiltonian has levels

$$E_n = \hbar\nu_c\left(n+\frac{1}{2}\right), \qquad n \geq 0.$$

The centre changes with k but the frequency does not. The factor $e^{\mathrm{i}ky}$ has constant modulus along the whole y-line and is not square-integrable there. Normalized planar states require square-integrable superpositions over k.

Exercise 4 (A coherent-state projection threshold). Recall the coherent-state overlap $|\langle\psi_\alpha,\psi_\beta\rangle|^2 = e^{-|\alpha-\beta|^2}$. For $\beta = \mathrm{i}\alpha$ find the probability of the rank-one projection onto ψ_β in state ψ_α. (i) Find the smallest $|\alpha|$ for which this probability is at most $1/10$. (ii) Compare with the projection onto $\psi_{-\alpha}$ at this same value of $|\alpha|$.

Solution. The Born probability is the squared coherent-state overlap. Since $|\alpha-\mathrm{i}\alpha|^2 = 2|\alpha|^2$, the probability and threshold are

$$p = e^{-2|\alpha|^2}, \qquad |\alpha|_{\min} = \sqrt{\frac{\log 10}{2}}.$$

For the opposite amplitude the squared separation is $4|\alpha|^2$. At the threshold just found its projection probability is

$$|\langle\psi_{-\alpha},\psi_\alpha\rangle|^2 = e^{-4|\alpha|^2} = \frac{1}{100}.$$